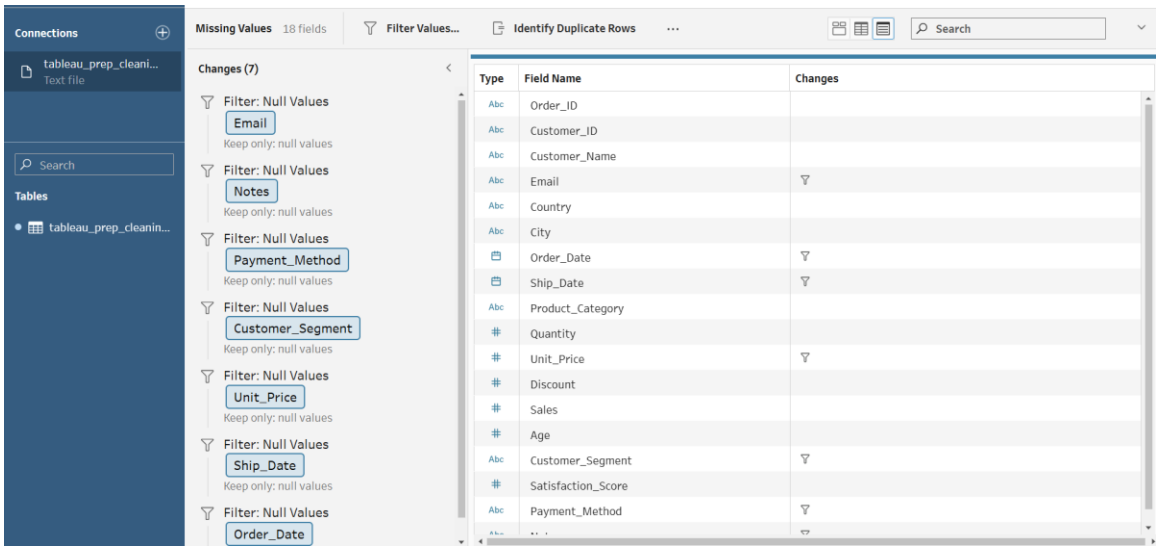
Data Cleaning and Profiling Report (Tableau Prep)

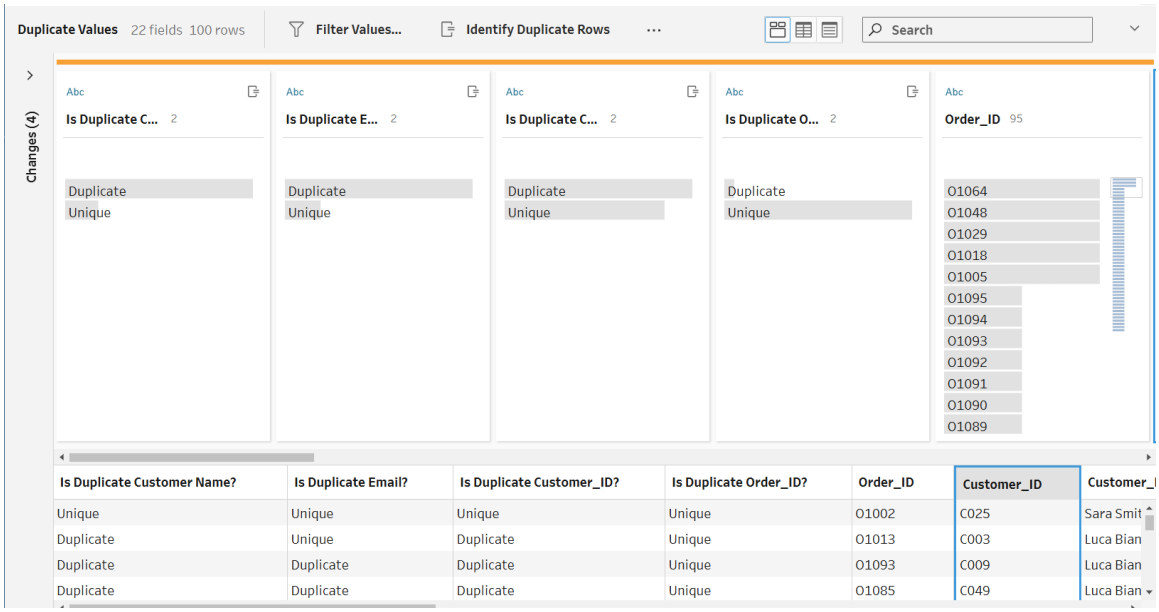
1. Dataset Description

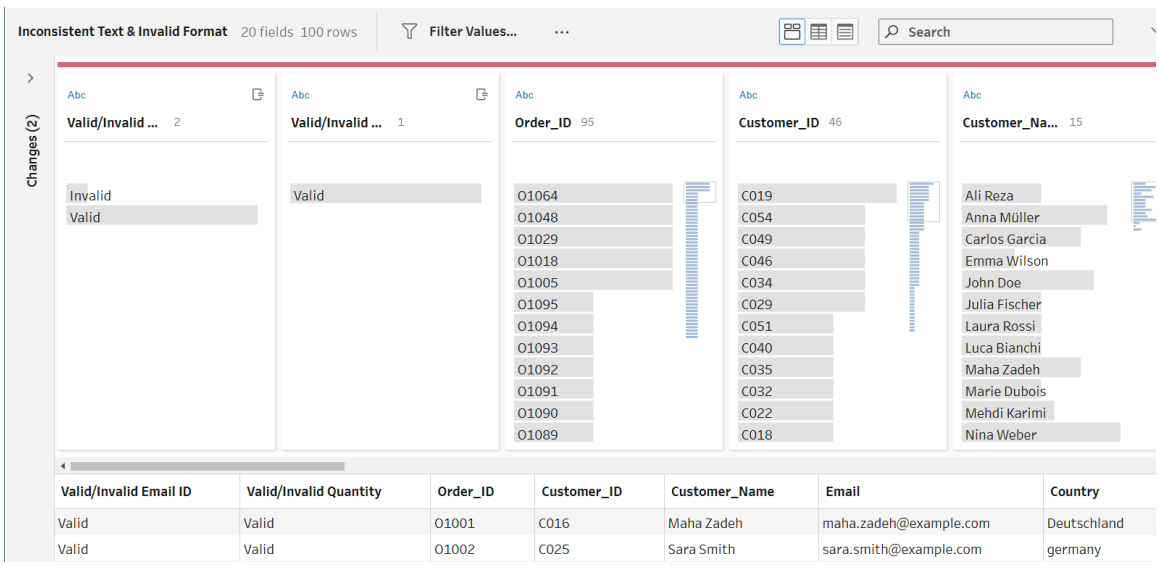
The dataset consists of transactional data including customer details, order information, pricing, and sales. It is used for analyzing customer behavior and sales performance.

2. Data Profiling

Initial profiling revealed missing values, inconsistent formats, duplicates, and invalid patterns in fields such as Email, Dates, and Quantity.

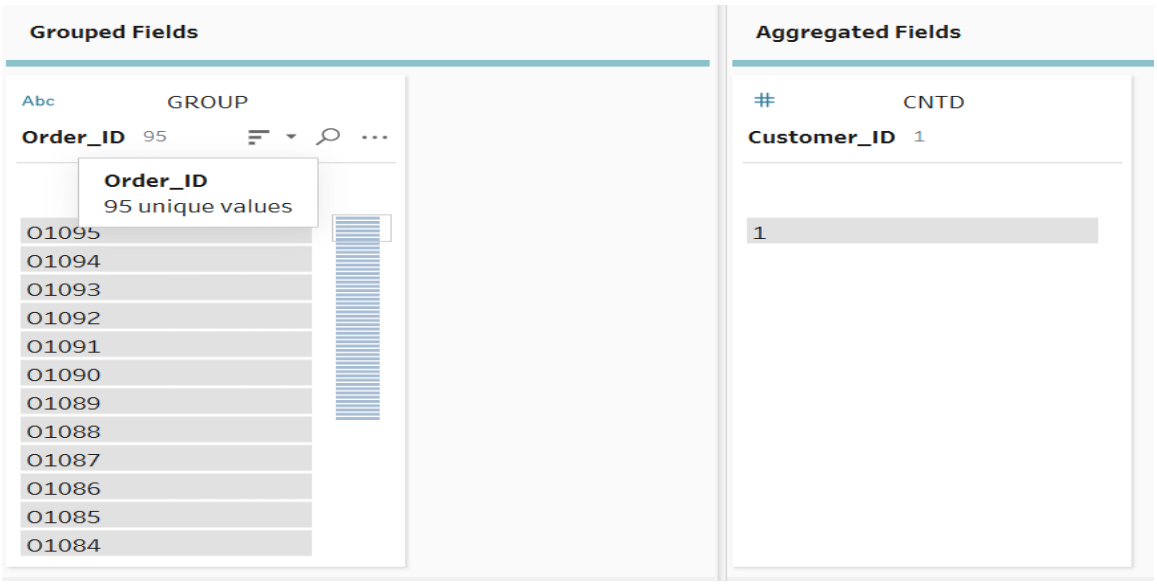


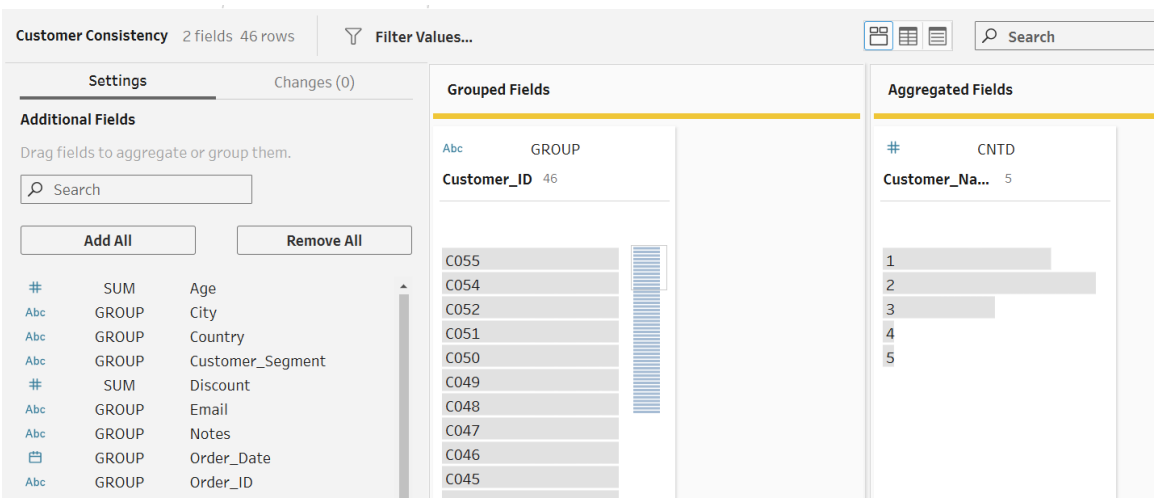




3. Structure Analysis

Order_ID was analyzed as a primary key. Inconsistencies were found where multiple Customer_IDs were linked to a single Order_ID. Relationships such as Customer_ID → Customer_Name were also validated.





Column Analysis:

• High Cardinality Columns:
- Order ID
- Customer ID
- Email

• Identifier Columns:
- Order ID → Primary identifier (should be unique)
- Customer ID, Email → Secondary identifiers

• Data Validation Findings:
- Dates:
  * Some invalid formats detected

  * Converted to Date → invalid values became NULL
- Emails:
  * Missing "@" or domain in some records
- Numeric Fields:
  * Missing values (NULL)
  * Negative values (e.g., Quantity)
  * Outliers (very high values)

4. Cleaning Rules

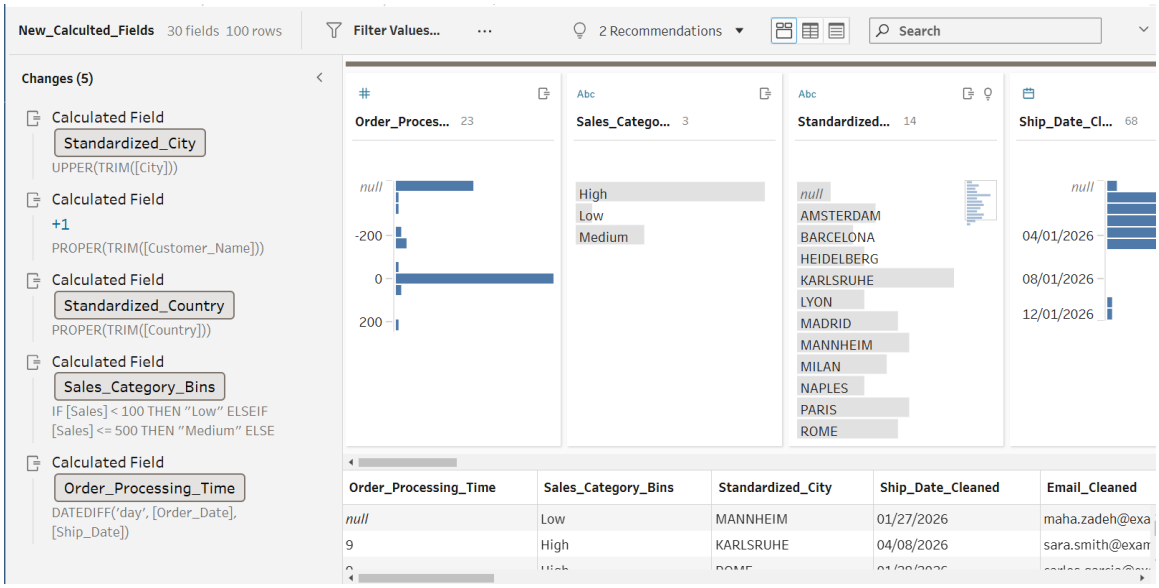
• Missing Values:
- Numeric fields replaced with median or 0
- Categorical fields replaced with "Unknown"
- Critical fields removed if missing

• Duplicates:
- Identified using Order_ID
- Retained most relevant record and removed duplicates

• Invalid Values:
- Negative quantities removed
- Invalid dates filtered out
- Incorrect email formats removed or corrected
- Text standardized using TRIM, UPPER, PROPER

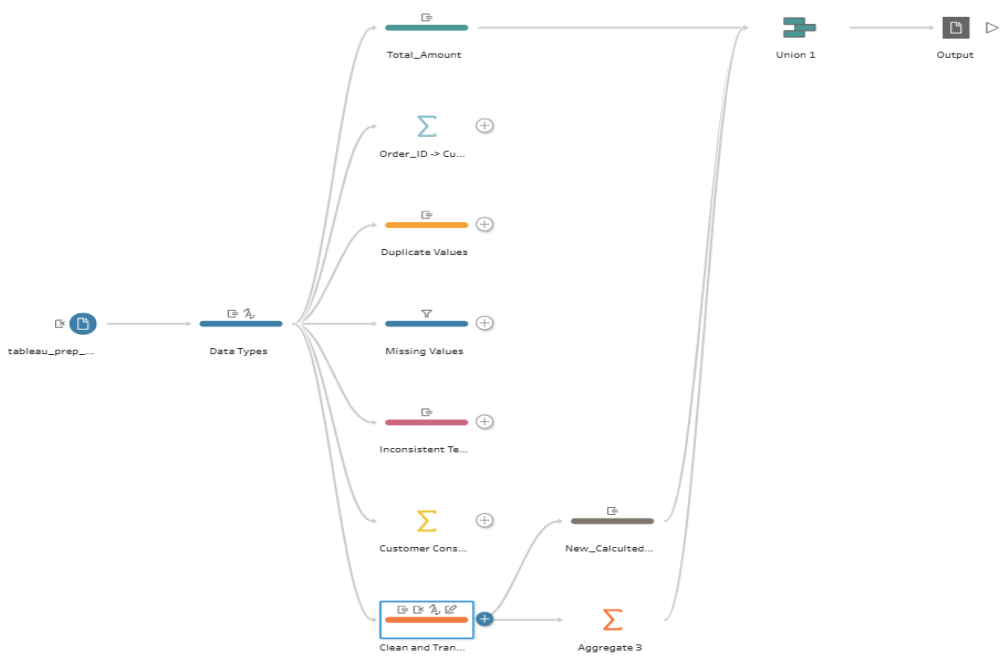
5. Data Transformation

• Text fields standardized using TRIM, UPPER, PROPER
• Dates converted to proper format
• Calculated field Total Amount created using Unit Price × Quantity and rounded
• Categories created for Sales (Low, Medium, High)



6. Validation

- Row counts compared before and after cleaning
- Missing values minimized
- Duplicate records removed
- Data types corrected
- Relationships validated (Order_ID uniqueness, Customer consistency)



7. Export

The cleaned dataset was exported in CSV and Hyper formats for further use in Tableau Desktop.

8. Pipeline Context

ETL approach was used where data was cleaned before loading. This ensures high data quality and is suitable for small datasets processed using Tableau Prep.

ETL:
- Profiling before loading
- High data quality

ELT:
- Profiling after loading
- More scalable but complex

ETL is preferred for this dataset.